AI Market Signal Research Assistant

End-to-End Implementation Plan

Investor-grade research-only system for automated market signals, opportunity analysis, evidence-backed scoring, and decision support

Document Purpose	Scope
Implementation blueprint	Build an AI assistant that continuously monitors market/search/competition signals, identifies low-competition high-reward opportunities, validates facts from APIs, and presents investor-style opportunity reports. It does not build the business idea; it only researches and ranks opportunities for human approval.

Refined Prompt

Create an end-to-end implementation plan for an AI research assistant that automatically generates market opportunity signals based on real market trends. The application should only perform in-depth research and present investor-style opportunity reports with factual evidence, measurable numbers, confidence levels, risk analysis, and source-backed reasoning. It should identify low-competition, high-reward ideas, but it must not proceed to build any idea. The user will review the research and decide whether an opportunity is worth building. The system must minimize incorrect data by validating facts, separating measured data from assumptions, and rejecting weak or unverifiable signals.

1. Executive Summary

The system is a research-only market intelligence assistant. Its job is to collect external market signals, validate them, cluster them into opportunities, score them, and present them like an investment memo. It must not act as a website builder, product builder, deployment bot, or content generator for execution. Its core output is a ranked list of opportunity signals and detailed investor-style reports.

- Primary output: investor-style opportunity memos with evidence, score, risks, numbers, and decision recommendation.
- Core principle: no invented metrics. Every number must be measured, calculated from measured data, or explicitly labeled as an estimate.
- Decision boundary: the app recommends whether an opportunity deserves investigation, but the human decides whether to build.
- Quality standard: prefer fewer high-confidence opportunities over many weak or hallucinated ideas.

What the app does	What the app does not do
Collect search, SERP, trend, competitor, CPC, and content gap data	Automatically build or deploy the selected opportunity
Generate market signals and opportunity scores	Invent search volume, CPC, ranking strength, TAM, or revenue numbers
Produce investor-grade opportunity reports	Write generic hype reports with unsupported claims

Flag confidence, data quality, missing data, and risk	Hide uncertainty or pretend weak evidence is strong
Let user approve, reject, watchlist, or revisit opportunities	Make final business decisions without user review

2. Product Requirements

Requirement	Implementation Detail	Priority
Automated signal generation	Scheduled jobs scan categories, keywords, SERPs, PAA, related searches, CPC, trend-like movement, competitor weakness, and content gaps.	Must have
Research-only scope	No website generation, no deployment, no automated purchasing, no publishing. Only research and decision support.	Must have
Investor-style reports	Every opportunity must include thesis, evidence, market demand, competition, moat, monetization, risks, confidence, and decision recommendation.	Must have
Fact-first data handling	Measured data, calculated scores, assumptions, and AI interpretation are stored separately.	Must have
Reject weak data	If evidence is insufficient, mark as Insufficient Evidence rather than producing a confident recommendation.	Must have
Human approval workflow	User can approve, reject, watchlist, compare, export, and add notes.	Must have
Opportunity monitoring	Watchlisted opportunities are rechecked periodically for trend changes.	Should have
Backtesting	System records old scores and compares changes over time.	Should have
Explainability	Each score includes formula inputs and reasons.	Must have
Data provenance	Every important claim links back to source, API, timestamp, and raw response summary.	Must have

3. High-Level Architecture

```
User Dashboard
  |
Research Request / Scheduled Auto-Discovery
  |
Signal Collection Layer
  |-- DataForSEO: keyword metrics, CPC, competition, volume
  |-- SerpAPI: Google SERP, PAA, related searches, local packs, news/images/video where relevant
  |-- GPT: seed expansion, clustering, report interpretation, but not metric invention
  |
```

Normalization + Validation Layer

|

Signal Detection Engine

|

Opportunity Clustering Engine

|

Scoring + Confidence Engine

|

Investor Memo Generator

|

Research Database + Evidence Store

|

Dashboard: ranked signals, memos, comparisons, watchlist, exports

Layer	Responsibility	Key Output
Input Layer	Accept seed topics, categories, region, language, thresholds, and schedule settings.	ResearchRun
Signal Collection Layer	Fetch keyword, SERP, competitor, related query, and market trend indicators.	RawSignal + NormalizedSignal
Validation Layer	Remove duplicates, detect contradictory data, score data freshness and completeness.	DataQualityScore
Signal Detection Engine	Identify rising demand, weak SERPs, CPC changes, new PAA questions, long-tail clusters, and underserved local gaps.	MarketSignal
Clustering Engine	Group related signals into investable opportunity themes.	OpportunityCluster
Scoring Engine	Calculate opportunity, confidence, risk, demand, commercial, competition, and execution scores.	ScoreBreakdown
Memo Generator	Create factual investor-style opportunity memos using measured data and clear assumptions.	OpportunityMemo
Decision Layer	Allow approve, reject, watchlist, compare, export, and revisit decisions.	DecisionRecord

4. Automated Market Signal Generation

Signals are early indicators that a niche may deserve deeper investigation. The app should generate signals automatically through scheduled category scans and seed expansion. A signal is not yet an opportunity. A signal becomes an opportunity only after validation, clustering, and scoring.

Signal Type	Trigger Logic	Example
Rising Demand Signal	Keyword volume, related searches, or query expansion indicates increasing demand compared with historical snapshots.	More queries around “AI resume checker for freshers”.
Low Competition Signal	SERP contains weak pages, forums, old posts, thin content, or non-specialized websites.	Reddit/Quora ranking for commercial query.

High Commercial Intent Signal	CPC, buyer-intent phrases, product modifiers, pricing queries, or comparison queries are present.	“best X tool”, “X price”, “X alternative”.
Content Gap Signal	Competitors lack calculators, tools, templates, local examples, comparison tables, or updated explanations.	No India-specific calculator exists.
Localization Gap Signal	Global content ranks in a region where local context matters.	US-focused tax/finance/software pages ranking in Indian queries.
Forum Pain Signal	Repeated questions in PAA, forums, Reddit/Quora, or community pages indicate unsolved demand.	Many users ask how to price 3D prints in India.
SERP Volatility Signal	Ranking pages change frequently or low-authority pages enter top results.	Weak domains appearing in top 10.
Tool Missing Signal	Search intent would be better served by a calculator, checklist, generator, comparison engine, or estimator.	Searchers want cost calculation, not a generic article.
Monetization Signal	CPC, affiliates, lead-gen value, SaaS potential, or high purchase intent is detected.	Keyword has CPC and products/services around it.
Risk Signal	Topic has legal, medical, financial, regulatory, or heavy brand dominance risk.	Health diagnosis topic with strong YMYL concerns.

5. Data Sources and Validation Rules

The system should treat API data as evidence, not truth by default. Every metric should have a source, timestamp, unit, region, and confidence level. If sources conflict, the report should show the conflict instead of hiding it.

Data Category	Primary Source	Validation Rule
Keyword volume	DataForSEO	Store exact region, language, match type, date, and source. If missing, mark unknown, not zero.
CPC and competition	DataForSEO	Use ranges and confidence bands. Avoid over-precise revenue predictions.
SERP competitors	SerpAPI	Capture top 10 or top 20 and classify content type. Re-run if API result is incomplete.
People Also Ask	SerpAPI	Treat as qualitative demand signal, not volume.
Related searches	SerpAPI	Use for seed expansion and long-tail detection. Validate with keyword metrics where possible.
Content weakness	SERP page metadata + optional fetch summary	Classify with rules first, GPT second. GPT must cite evidence from SERP data.
Trend movement	Historical snapshots inside own database	Compare current run with prior runs. Avoid claiming growth without historical baseline.
Revenue potential	Calculated estimate	Label as estimate and show formula. Do not present as fact.

Data Quality State	Meaning	System Action
Verified	Metric comes from reliable API response and passes validation.	Use in scoring.

Partial	Some values are missing or region/language coverage is incomplete.	Use with reduced confidence.
Conflicting	Multiple sources or runs disagree strongly.	Flag conflict and reduce confidence.
Stale	Data is older than threshold.	Request refresh or lower confidence.
Insufficient Evidence	Not enough measurable support.	Do not generate strong recommendation.

6. “No Wrong Data” Policy

No system can guarantee perfect market data, but this app can be designed to avoid confidently presenting incorrect or unsupported information. It should prefer conservative reporting, confidence labels, and evidence-first scoring.

Rule	Implementation
Never invent metrics	GPT cannot create search volume, CPC, competition, growth rate, revenue, or TAM unless computed by code from available inputs.
Unknown is allowed	Missing data must be displayed as Unknown or Not Available, never silently replaced with zero.
Separate facts from assumptions	Reports have separate sections for measured facts, calculated estimates, assumptions, and AI interpretation.
Use validation gates	A signal cannot become an opportunity unless minimum evidence thresholds are met.
Confidence scoring	Every opportunity receives a confidence score based on data completeness, freshness, consistency, and source quality.
Contradiction handling	When data conflicts, show both values and reduce confidence.
Audit trail	Every score can be traced back to raw inputs and formulas.
AI output validation	All GPT JSON output is schema-validated. Invalid output is retried or rejected.
No hype language	Investor memo tone should be analytical, not promotional.

7. Signal-to-Opportunity Lifecycle

Stage 1: Raw Signal

Example: PAA and related searches show many questions around a topic.

Stage 2: Validated Signal

Keyword metrics and SERP analysis confirm enough measurable demand and weak competition.

Stage 3: Opportunity Cluster

Multiple validated signals group into a coherent niche opportunity.

Stage 4: Scored Opportunity

System calculates demand, competition, commercial, content gap, execution, risk, and confidence scores.

Stage 5: Investor Memo

System presents the opportunity with evidence, assumptions, risks, and recommendation.

Stage 6: Human Decision

User marks: Approve for build consideration, Reject, Watchlist, or Needs More Research.

8. Scoring Engine

The scoring engine should be deterministic and code-driven. GPT may explain the result, but the numeric score must be calculated by the backend.

Score	Weight	Input Signals
Demand Score	25%	Total search volume, long-tail volume, number of related keywords, regional demand, historical growth if available.
Competition Advantage Score	25%	Keyword difficulty, weak SERP ratio, forum ranking ratio, outdated pages, non-local competitors, thin content.
Commercial Score	20%	CPC, buyer intent modifiers, affiliate/SaaS/lead-gen potential, product/service value.
Content Gap Score	15%	Missing tools, templates, calculators, local content, comparison pages, examples, UX quality.
Execution Feasibility Score	10%	MVP complexity, content difficulty, data availability, cost, maintenance burden.
Trend Signal Score	5%	New/accelerating related queries, recurring signal growth, watchlist momentum.

```
Base Opportunity Score =
  Demand Score * 0.25 +
  Competition Advantage Score * 0.25 +
  Commercial Score * 0.20 +
  Content Gap Score * 0.15 +
  Execution Feasibility Score * 0.10 +
  Trend Signal Score * 0.05

Final Opportunity Score = Base Opportunity Score - Risk Penalty

Confidence Score = function(
  data_completeness,
  source_quality,
  freshness,
  consistency,
  sample_size,
  validation_pass_rate
)
```

Risk Penalty	Typical Penalty
YMYL or regulated domain	5 to 25 points
Strong brand-dominated SERP	5 to 20 points

Low data confidence	5 to 30 points
Seasonal or temporary demand	3 to 15 points
High execution complexity	3 to 15 points
No clear monetization	5 to 25 points

9. Validation Gates and Thresholds

Gate	Default Rule	Result if Failed
Minimum demand	Cluster should have measurable search demand or repeated validated signals.	Mark as weak demand.
Minimum evidence count	At least 3 independent evidence points across keyword/SERP/PAA/related searches/competitor gap.	Mark as insufficient evidence.
SERP completeness	At least top 10 results captured for core keyword.	Retry or lower confidence.
Metrics completeness	Search volume, CPC, or competition present for core keywords where API supports it.	Use partial scoring only.
AI interpretation check	GPT output must reference provided facts only.	Reject output and retry with stricter prompt.
Opportunity coherence	Cluster keywords must share intent and target user.	Split cluster or discard.

10. Investor-Style Dashboard

Page	Purpose	Key Components
Home	Executive overview of latest signals and watchlist changes.	Top signals, score changes, category heatmap, pending decisions.
Signal Feed	Raw and validated signals detected automatically.	Signal type, evidence, region, category, confidence, status.
Opportunities	Ranked investable opportunity clusters.	Score, confidence, risk, demand, competition, monetization, action.
Opportunity Memo	Full investor-grade report.	Thesis, evidence, numbers, risks, competitors, recommendation.
Compare	Compare multiple opportunities side by side.	Score matrix, risk matrix, execution complexity, monetization options.
Watchlist	Track opportunities over time.	Score changes, new SERP changes, volume movement, alerts.
Research Runs	Audit log of each scan.	Inputs, status, API counts, outputs, failed validations.
Settings	Configure categories, regions, thresholds, scoring weights, schedules.	Research scope and scoring preferences.

11. Investor Memo Output Template

```
{
  "memo_type": "Market Opportunity Research Memo",
  "title": "",
  "decision_status": "New | Watchlist | Approved for Build Consideration | Rejected | Needs More Research",
  "one_line_thesis": "",
```

"target_region": "",
"target_audience": [],
"category": "",
"opportunity_score": 0,
"confidence_score": 0,
"risk_score": 0,
"data_quality_state": "Verified | Partial | Conflicting | Stale | Insufficient Evidence",
"measured_facts": {
  "total_monthly_search_volume": 0,
  "average_cpc": "",
  "average_keyword_difficulty": 0,
  "serp_weakness_ratio": 0,
  "forum_result_count": 0,
  "paa_questions": [],
  "related_searches": []
},
"calculated_estimates": {
  "traffic_capture_scenarios": [],
  "monetization_scenarios": [],
  "estimated_mvp_effort": ""
},
"assumptions": [],
"market_problem": "",
"why_this_signal_exists": "",
"evidence_summary": [],
"competitor_landscape": [],
"content_and_product_gaps": [],
"possible_solution_angles": [],
"monetization_paths": [],
"risks_and_unknowns": [],
"recommended_next_research": [],
"final_recommendation": "Investigate | Watchlist | Reject | Strong Candidate"
}

12. Backend API Design

Route	Method	Purpose
/api/research/start	POST	Start manual research for a seed topic.
/api/research/auto-discover	POST	Start automated category-based signal discovery.
/api/research/runs	GET	List previous research runs.
/api/research/runs/:id	GET	Get full research run details.
/api/signals	GET	List generated signals with filters.
/api/signals/:id	GET	View signal evidence and validation state.
/api/opportunities	GET	List ranked opportunity clusters.
/api/opportunities/:id	GET	View investor memo and score

		breakdown.
/api/opportunities/:id/watchlist	POST	Add opportunity to watchlist.
/api/opportunities/:id/approve	POST	Mark as approved for build consideration.
/api/opportunities/:id/reject	POST	Reject opportunity with reason.
/api/opportunities/:id/request-more-research	POST	Queue deeper validation.
/api/compare	POST	Compare multiple opportunities.
/api/settings/scoring	GET/POST	Read or update scoring weights.

13. Database Schema

Core models:

ResearchRun

- id, mode, seedTopic, category, region, language, status, startedAt, completedAt, createdAt, updatedAt

RawApiResponse

- id, researchRunId, provider, endpoint, requestHash, responseSummary, status, fetchedAt

KeywordMetric

- id, researchRunId, keyword, region, language, searchVolume, cpcLow, cpcHigh, competition, difficulty, intent, source, fetchedAt

SerpResult

- id, researchRunId, keyword, rank, title, url, domain, snippet, contentType, serpFeatureType, source, fetchedAt

MarketSignal

- id, researchRunId, signalType, title, description, category, region, language, evidenceJson, confidenceScore, validationState, createdAt

OpportunityCluster

- id, researchRunId, title, category, region, primaryIntent, totalVolume, avgCpc, avgDifficulty, signalCount, validationState, createdAt

ScoreBreakdown

- id, opportunityClusterId, demandScore, competitionAdvantageScore, commercialScore, contentGapScore, executionScore, trendSignalScore, riskPenalty, finalScore, confidenceScore, riskScore, formulaVersion

OpportunityMemo

- id, opportunityClusterId, memoJson, measuredFactsJson, calculatedEstimatesJson, assumptionsJson, recommendation, createdAt

DecisionRecord

- id, opportunityMemoId, decision, notes, decidedAt

WatchlistItem

- id, opportunityClusterId, status, nextReviewAt, lastScore, currentScore, createdAt, updatedAt

EvidenceItem

- id, parentType, parentId, sourceType, sourceName, metricName, value, unit, url, timestamp, confidence, notes

ApiUsageLog

- id, researchRunId, provider, endpoint, requestCount, status, createdAt

14. Core Workflows

14.1 Automated Daily Signal Discovery

1. Scheduler starts daily category scan.
2. GPT generates seed candidates for enabled categories.
3. System fetches keyword metrics for seed candidates.
4. System removes candidates below minimum evidence threshold.
5. System fetches SERP data for surviving candidates.
6. Signal engine detects rising demand, weak competition, commercial intent, and gaps.
7. Related signals are clustered into opportunities.
8. Scoring engine calculates scores.
9. Memo generator creates investor-style reports for top opportunities only.
10. Dashboard shows new signals, upgraded signals, downgraded signals, and rejected signals.

14.2 Manual Deep Research

1. User enters a topic like "3D printing India".
2. System expands topic into related seeds.
3. System validates actual search demand.
4. System analyzes top SERPs.
5. System clusters opportunities.
6. System generates score breakdown.
7. System creates investor memo.
8. User marks: approve, reject, watchlist, or needs more research.

14.3 Watchlist Monitoring

1. User watchlists an opportunity.
2. System rechecks core keywords and SERPs on schedule.
3. System compares new data with previous snapshots.
4. If score changes materially, system creates an alert.
5. Memo gets updated with a score movement section.

15. GPT Usage Boundaries

GPT Task	Allowed	Not Allowed
Seed expansion	Generate candidate keywords and angles.	Claim demand exists before API validation.
Clustering	Group validated keywords/signals by intent.	Merge unrelated topics just because they sound similar.
SERP interpretation	Summarize observed weaknesses from	Invent page quality or domain metrics

	provided SERP data.	not provided.
Memo writing	Create readable investor-style narrative from evidence.	Invent facts, cite nonexistent data, or overstate certainty.
Risk analysis	Identify risks from available facts and known categories.	Make legal/medical/financial conclusions as expert advice.

16. Codex Implementation Prompt

Use this prompt with Codex to build the application. Keep the scope research-only.

```
Build a production-ready AI Market Signal Research Assistant.

Core scope:
- The app only researches opportunities and presents investor-grade reports.
- It must not build, deploy, publish, or execute the business idea.
- It must automatically generate market signals from search, SERP, trend-like, competitor, and content gap data.
- It must use factual data only. No invented metrics.
- It must separate measured data, calculated estimates, assumptions, and AI interpretation.
- It must reject or downgrade opportunities with insufficient evidence.

Build modules:
1. Dashboard with Signal Feed, Opportunities, Opportunity Memo, Compare, Watchlist, Research Runs, Settings.
2. Data integrations for keyword metrics and Google SERP collection.
3. Normalization layer for keyword metrics, SERP results, PAA, related searches, competitor classifications.
4. Validation layer with data quality states: Verified, Partial, Conflicting, Stale, Insufficient Evidence.
5. Automated signal detection engine.
6. Opportunity clustering engine.
7. Deterministic scoring engine with formula versioning.
8. GPT memo generator with schema validation.
9. Decision workflow: approve for build consideration, reject, watchlist, request more research.
10. Watchlist monitoring with score change alerts.
11. Audit trail for every score and claim.
12. Tests for scoring, validation, clustering, API normalization, and invalid AI output.

Acceptance criteria:
- A user can run manual research for a seed topic.
- A scheduled auto-discovery job can generate market signals automatically.
- Signals are validated before becoming opportunities.
- Opportunities are scored with evidence and confidence.
- Investor memos contain measured facts, calculated estimates, assumptions, risks, and recommendation.
- If data is weak, the system says Insufficient Evidence instead of creating a false confident report.
- The app provides no build/deploy automation for the selected idea.
```

17. Internal AI Prompt Templates

17.1 Seed Candidate Prompt

```
You are a market research assistant.
Generate seed candidates for research only.
Do not claim demand exists.
Return ideas that must be validated by keyword and SERP APIs.

Input:
Category: {{category}}
Region: {{region}}
Language: {{language}}

Return JSON:
{
  "seed_candidates": [
    {"seed": "", "reason_to_validate": "", "possible_user_problem": "", "expected_intent": ""}
  ]
}
```

17.2 Signal Interpretation Prompt

```
You are an evidence-based market signal analyst.
Use only the provided data.
Do not create numbers.
If evidence is weak, say so.

Input:
Keyword metrics: {{keyword_metrics}}
SERP results: {{serp_results}}
PAA/related searches: {{paa_related}}
Historical snapshots: {{historical_snapshots}}

Return JSON:
{
  "signals": [
    {
      "signal_type": "",
      "description": "",
      "evidence": [],
      "confidence": 0,
      "limitations": []
    }
  ]
}
```

17.3 Investor Memo Prompt

```
You are an investor-style market research analyst.
Create a factual opportunity memo using only provided evidence.
Separate facts, estimates, assumptions, and recommendations.
Do not invent numbers.
Do not write hype.

Input:
Opportunity cluster: {{cluster}}
Measured facts: {{measured_facts}}
Score breakdown: {{score_breakdown}}
Evidence items: {{evidence_items}}
Risks: {{risks}}

Return JSON matching the OpportunityMemo schema.
```

18. Recommended Tech Stack

Area	Recommended Stack	Reason
Frontend	Next.js, TypeScript, Tailwind, shadcn/ui, Recharts	Fast dashboard development with good tables and charts.
Backend	Next.js API routes or FastAPI services	Keep MVP simple; split into services later if needed.
Database	PostgreSQL + Prisma	Good relational model for runs, metrics, signals, evidence, and memos.
Jobs	BullMQ, Inngest, Trigger.dev, or Cloudflare Queues	Scheduled scans and background research runs.
AI	OpenAI model with structured JSON outputs	Used for interpretation, clustering, and memo writing only.
Charts	Recharts	Score breakdown, trend movement, category heatmaps.
Testing	Vitest/Jest, Playwright, mocked API providers	Ensure scoring and validation reliability.

19. Implementation Phases

Phase	Build	Done When
Phase 1: Research-only MVP foundation	Next.js app, database, dashboard shell, research run creation, opportunity pages.	User can create a research run and view empty-state pages.
Phase 2: Data collection	Keyword API integration, SERP API integration, normalized models, raw response storage.	Seed research fetches and stores real keyword and SERP data.
Phase 3: Signal engine	Signal types, validation gates, confidence scoring, signal feed UI.	System generates validated signals from data.
Phase 4: Clustering and scoring	Opportunity clustering, deterministic score formulas, risk penalties.	System ranks opportunities with score breakdown.
Phase 5: Investor memo generator	GPT prompts, schema validation, evidence separation, memo UI.	Each opportunity has an investor-style memo.
Phase 6: Auto-discovery	Scheduled scans by category, thresholds, category settings.	System automatically discovers signals without manual seed input.

Phase 7: Watchlist and compare	Watchlist rechecks, score history, compare page.	User can track and compare opportunities over time.
Phase 8: Hardening	Tests, mocks, retries, data quality checks, UI polish.	System is reliable for daily use.

20. Testing Strategy

Test Area	Examples
Scoring engine	Same inputs always return same score; missing values reduce confidence; risk penalty applies correctly.
Validation layer	Unknown values are not treated as zero; conflicting data is flagged; stale data is downgraded.
API normalization	DataForSEO and SerpAPI responses map correctly into internal models.
AI schema validation	Invalid JSON is retried; unsupported claims are rejected; missing measured facts are not fabricated.
Signal generation	Forum-heavy SERP creates weak competition signal; missing tool creates content gap signal.
Memo generation	Report separates measured facts, estimates, assumptions, and recommendations.
UI	Opportunity list filters work; memo page shows score and evidence; watchlist updates appear.

21. Example Investor Memo Summary

Opportunity: 3D Printing Cost Calculator for India

Decision Status: New

Recommendation: Strong Candidate for Further Investigation

Opportunity Score: 81/100

Confidence Score: 74/100

Risk Score: 22/100

Thesis:

Indian 3D printing hobbyists and small print farms need localized cost estimation tools for filament, electricity, printer depreciation, failed prints, and profit margin. Existing content is mostly generic, non-local, or incomplete.

Measured Facts:

- Keyword metrics: Stored from keyword API response.
- SERP competitors: Stored from top Google results.
- PAA/related searches: Stored from SERP API.

Why it may be attractive:

- Search intent is practical and tool-oriented.
- A calculator can serve the intent better than a generic blog.
- Monetization paths include ads, affiliate links, lead generation, and premium templates.

Risks:

- Demand may be niche.

- Affiliate revenue depends on partner availability.

- Search volume must be validated across multiple related long-tail queries.

Next Research:

- Validate 20 related keywords.

- Check SERP weakness for top 5 queries.

- Compare against existing calculators.

- Estimate traffic scenarios conservatively.

22. Final Acceptance Criteria

- The app automatically generates market signals from scheduled scans.
- Signals are validated before becoming opportunities.
- The app is research-only and does not build or deploy selected ideas.
- Every opportunity has measured facts, calculated estimates, assumptions, risks, and recommendation.
- No search volume, CPC, competition, growth, revenue, or TAM is invented by GPT.
- Unknown data is displayed as unknown, not zero.
- Every score has an explainable breakdown and formula version.
- Weak or incomplete evidence results in Insufficient Evidence or reduced confidence.
- The dashboard presents the information like an investor decision memo.
- The user can approve, reject, watchlist, compare, or request more research.
- Historical snapshots allow trend movement and score changes to be tracked over time.
- Tests cover scoring, validation, normalization, signal detection, and AI output validation.

23. Build Order for Codex

1. Create app shell and dashboard navigation.
2. Create database models for research runs, keyword metrics, SERP results, signals, clusters, scores, memos, evidence, decisions, and watchlist.
3. Implement manual research run creation.
4. Implement keyword and SERP data providers with mocked providers for testing.
5. Implement normalization and validation layer.
6. Implement market signal generation rules.
7. Implement opportunity clustering.
8. Implement deterministic scoring engine.
9. Implement investor memo generator with strict JSON schema validation.
10. Build Signal Feed, Opportunities list, Opportunity Memo, Compare, Watchlist, and Research Runs pages.
11. Implement scheduled auto-discovery by category.
12. Add score history and watchlist monitoring.
13. Add tests and harden failure handling.